Isolation (ACID Property)

1. Overview

Isolation means concurrent transactions should execute **without interfering with each other**.

```None
A transaction’s intermediate changes

must remain invisible to others

until commit.
```

Even if many users run transactions simultaneously, each transaction should behave **as if it were running alone**.

2. Simple Meaning

During execution:

```None
Uncommitted changes = hidden

Committed changes = visible
```

No other transaction should read partial results.

3. Banking Example

Transfer ₹50 from Account A to Account B.

Steps:

```None
1. Deduct ₹50 from A

2. Add ₹50 to B

3. Commit
```

Suppose another transaction reads data after step 1 but before step 2.

It may see:

```None
A = 450

B = 300

Total = 750
```

This is temporary intermediate state.

Isolation prevents others from seeing this incomplete result.

4. Correct Behavior

Other transactions should either see:

Before Commit

```None
A = 500

B = 300
```

or

After Commit

```None
A = 450

B = 350
```

But not partial in-between state.

5. Why Isolation Matters

Without isolation:

- Dirty reads
- Lost updates
- Double booking
- Incorrect balances
- Duplicate stock sales
- Race conditions

6. Example: Ticket Booking

Two users try to buy last seat simultaneously.

Without isolation:

```None
User A reads seat available

User B reads seat available

Both purchase
```

Overselling occurs.

7. Isolation Levels (Important Interview Topic)

Most databases support levels:

Level	Prevents
Read Uncommitted	Lowest safety
Read Committed	Dirty reads
Repeatable Read	Dirty + non-repeatable reads
Serializable	Full isolation

Examples in PostgreSQL and MySQL

8. Common Concurrency Problems

A. Dirty Read

Read uncommitted data.

B. Non-Repeatable Read

Same row read twice gives different value.

C. Phantom Read

Repeated query returns extra rows.

D. Lost Update

Two transactions overwrite each other.

9. How Databases Implement Isolation

Common techniques:

- Row locks
- Table locks
- MVCC (Multi-Version Concurrency Control)
- Timestamps
- Serializable scheduling

Example: PostgreSQL uses MVCC.

10. Isolation vs Atomicity

Property	Meaning
Atomicity	All or nothing
Isolation	Concurrent transactions don't interfere

11. Isolation vs Consistency

Property	Meaning
Isolation	Prevents concurrent interference
Consistency	Preserves valid rules

Isolation often helps maintain consistency.

12. Real SQL Example

```sql
BEGIN;


UPDATE accounts

SET balance = balance - 50

WHERE id='A';


UPDATE accounts

SET balance = balance + 50

WHERE id='B';


COMMIT;
```

Other sessions should not see half-finished state.

13. Performance Trade-off

Higher isolation:

Safer
More locking

Lower concurrency
More waiting

Choose based on workload.

14. Interview Answer Example

Q: What is isolation?

> Isolation ensures simultaneous transactions do not observe each other's intermediate changes, so each behaves as if executed independently.

15. Short Revision Notes

```None
Isolation =

Concurrent txns should not interfere



Hidden until commit



Prevents:

dirty read

lost update

double booking
```

16. One-Line Summary

Isolation ensures concurrent transactions execute safely by hiding uncommitted changes and preventing interference between simultaneous operations.